

F I N A L Y E A R P R O J E C T R E P O R T

H O N E Y P O T

D E F E N S E S Y S T E M

Production-Grade Network Honeypot with Real-Time Threat Intelligence

*A Multi-Protocol Deception Platform with Attacker Profiling, MITRE ATT&CK Mapping, and
Automated Deployment*

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ABSTRACT

Project Summary

This Final Year Project presents the HoneyPot Defense System, a production-grade multi-protocol network honeypot platform implemented in Node.js. The system deploys four deception services — SSH, HTTP, FTP, and Telnet — that emulate real production servers to attract, capture, and analyze attacker behavior in real time. It features deep attacker profiling with OS fingerprinting, MITRE ATT&CK TTP mapping, behavioral scoring, credential pattern analysis, and a glassmorphism-styled real-time monitoring dashboard with WebSocket push updates.

The platform addresses the critical cybersecurity challenge of early threat detection by deploying **interactive deception services** that convincingly mimic vulnerable infrastructure. Unlike passive network monitoring tools, the HoneyPot Defense System actively engages attackers, records their complete session data including commands executed, credentials attempted, tools used, and attack patterns employed — building comprehensive threat intelligence profiles for each unique attacker IP.

Key innovations include: a **unified EventEmitter-based architecture** connecting all services through a central message bus; an **AttackerProfiler engine** that performs real-time OS detection from 24 User-Agent patterns and 12 SSH client signatures, maps attacks to 18+ MITRE ATT&CK technique IDs, and scores attacker behavior on a 0-100 scale; a **PCAP synthesis engine** that reconstructs TCP sessions for Wireshark-compatible analysis; and a comprehensive **automated deployment system** supporting 8+ Linux distributions with full error recovery, systemd integration, firewall auto-configuration, and rollback capability.

The monitoring dashboard provides 10 specialized views including live attack feeds, attacker intelligence profiles, credential analysis, PCAP stream inspection with hex dump visualization, and professional threat reports with print-ready formatting. The system integrates with ClickHouse for analytics, Suricata IDS for signature-based detection, and external threat intelligence APIs (VirusTotal, AbuseIPDB) for IoC enrichment. The production codebase

comprises **~4,000 lines** across 10 Node.js modules and a 1,505-line single-page dashboard application.

Keywords: *Network Honeypot, Deception Technology, Threat Intelligence, MITRE ATT&CK, Attacker Profiling, Real-Time Monitoring, SSH/HTTP/FTP/Telnet, Behavioral Analysis, PCAP Analysis, Intrusion Detection, Node.js, WebSocket, Automated Deployment*

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

HoneyPot Defense System is a production-grade, multi-protocol network honeypot platform built entirely in Node.js. It deploys four interactive deception services — SSH (port 2222), HTTP (port 8080), FTP (port 2121), and Telnet (port 2323) — that emulate real production servers to attract and trap attackers. Every interaction is captured, analyzed, and presented through a real-time glassmorphism-styled monitoring dashboard with deep threat intelligence capabilities.

The system is designed with forward-looking modularity, enabling seamless integration into enterprise SOC workflows. All services communicate through a central EventEmitter bus, producing unified attack events that feed into persistent LevelDB storage, ClickHouse analytics, attacker profiling, and real-time WebSocket-powered dashboard updates.

1.2 Motivation and Need of the Hour

The cybersecurity landscape in 2025-2026 faces unprecedented challenges. According to **IBM's Cost of a Data Breach Report 2024**, the average dwell time for a breach is **204 days**, and the average cost has reached **\$4.88 million**. Organizations need early warning systems that detect threats before they reach production infrastructure.

Key industry challenges driving this project include:

- **Volume Overload:** Security Operations Centers (SOCs) receive thousands of alerts daily. Honeypots provide high-fidelity alerts with near-zero false positives — every interaction is inherently suspicious.
- **Threat Intelligence Gap:** Traditional firewalls and IDS block attacks but capture minimal attacker behavior data. Honeypots record complete session transcripts, credential dictionaries, tool signatures, and attack sequences.
- **Active Defense Shortage:** Most security tools are passive monitors. Deception technology actively wastes attacker time and resources while gathering intelligence.

- **Skill Shortage:** There is a global shortage of 3.4 million cybersecurity professionals (ISC² 2023). Automated profiling and MITRE ATT&CK mapping reduces the expertise needed to interpret attack data.
- **Dwell Time Reduction:** Early detection through honeypot alerts can reduce breach dwell time from months to minutes.

1.3 Problem Statement

Existing honeypot solutions suffer from significant limitations:

- **Single Protocol:** Most honeypots (Cowrie, Dionaea, Kippo) focus on one or two protocols, missing multi-vector attacks.
- **No Attacker Intelligence:** Traditional honeypots log raw events but don't profile attackers — no OS detection, tool identification, or behavioral scoring.
- **No MITRE Mapping:** Attack events aren't classified against the MITRE ATT&CK framework, making it difficult to understand attacker TTPs.
- **Deployment Complexity:** Most honeypots require manual configuration, dependency installation, and service management with no automated deployment.
- **Poor Visualization:** CLI-only interfaces or basic web UIs that don't provide professional threat reporting or real-time monitoring.

Gap Identified

There is a clear need for a unified, multi-protocol honeypot platform that combines interactive deception services, deep attacker intelligence with automated MITRE ATT&CK mapping, real-time visualization, and one-command production deployment — all in a single, open-source package.

1.4 Proposed Solution

The HoneyPot Defense System addresses every identified gap through a **four-layer architecture**:

LAYER	COMPONENTS	FUNCTION
Deception	SSH, HTTP, FTP, Telnet Honeypots	Attract and engage attackers across 4 protocols

LAYER	COMPONENTS	FUNCTION
Intelligence	AttackerProfiler, IDS Engine, OSINT Engine	OS detection, tool identification, MITRE mapping, behavioral scoring
Storage	LevelDB, ClickHouse, PCAP Engine	Persistent event storage, analytics, packet capture
Presentation	Express API, WebSocket, Dashboard	Real-time monitoring, reporting, data export

1.5 Objectives

The primary objectives of this project are:

- Design and implement a multi-protocol honeypot system supporting SSH, HTTP, FTP, and Telnet services with convincing emulation.
- Build an attacker profiling engine with OS fingerprinting, tool detection (24+ tools), and automated MITRE ATT&CK TTP mapping (18+ techniques).
- Develop a real-time monitoring dashboard with glassmorphism UI, 10 specialized views, WebSocket live updates, and professional report generation.
- Implement a PCAP synthesis engine for Wireshark-compatible session reconstruction.
- Create a credential pattern analyzer that classifies attack strategies (password spray, dictionary attack, username enumeration).
- Integrate with external threat intelligence APIs (VirusTotal, AbuseIPDB) and Suricata IDS for enriched detection.
- Build a fully automated deployment system supporting 8+ Linux distributions with error recovery, rollback, and systemd integration.
- Produce persistent storage with LevelDB and optional ClickHouse analytics for long-term attack trend analysis.

CHAPTER 2: LITERATURE REVIEW

2.1 Evolution of Honeypots

First Generation honeypots were simple port listeners that logged connection attempts. Tools like Honeyd (2003) could simulate network topologies but provided no interaction — attackers received no response, limiting intelligence gathered.

Second Generation honeypots introduced protocol emulation. Kippo (2009) and its successor Cowrie (2015) provided interactive SSH sessions with fake filesystems. Dionaea focused on malware collection via service emulation. These tools were single-protocol and required manual analysis.

Third Generation deception platforms — the current frontier — combine multiple protocols, automated analysis, and real-time intelligence. Commercial platforms like Attivo Networks and Illusive Networks offer enterprise-grade deception but are prohibitively expensive. **The HoneyPot Defense System represents this third generation** in the open-source space, integrating multi-protocol deception with automated profiling and MITRE mapping.

2.2 Existing Tools and Their Limitations

TOOL	STRENGTHS	LIMITATIONS	OUR ADVANTAGE
Cowrie	Mature SSH/Telnet honeypot	No HTTP/FTP, no profiling, no dashboard	4 protocols + full intelligence
Dionaea	Multi-protocol, malware capture	Complex setup, no real-time UI, no MITRE	One-command deploy + MITRE mapping
T-Pot	Multi-honeypot platform	Very heavy (30+ containers), complex ops	Lightweight single-process Node.js
HoneyD	Network topology simulation	No interaction, abandoned since 2007	Active development, interactive services
OpenCanary	Easy deployment, alerting	Low interaction, limited protocols	Medium-high interaction, deep intel
Artillery	Simple Python honeypot	Port monitoring only, no emulation	Full protocol emulation + profiling

2.3 MITRE ATT&CK Framework in Deception

The **MITRE ATT&CK** (Adversarial Tactics, Techniques, and Common Knowledge) framework catalogs over 200 techniques across 14 tactics. Our system maps honeypot events directly to ATT&CK technique IDs, enabling automated threat classification. The profiler maps 18+ technique IDs including T1110 (Brute Force), T1078 (Valid Accounts), T1059 (Command and Scripting Interpreter), T1083 (File and Directory Discovery), T1005 (Data from Local System), and T1105 (Ingress Tool Transfer).

2.4 Modern Deception Technology

Modern deception technology operates on the principle that any interaction with a honeypot is inherently malicious — eliminating the false positive problem that plagues traditional IDS/IPS systems. By combining medium-interaction emulation with automated intelligence extraction, honeypots provide SOC analysts with high-confidence, actionable threat data that requires no signature updates or rule tuning.

CHAPTER 3: SYSTEM ARCHITECTURE

3.1 EventEmitter Bus Architecture

The system uses Node.js's built-in **EventEmitter** as a central message bus (maxListeners=50). All honeypot services and the monitoring server subscribe to a shared bus instance. This pattern provides:

- **Loose coupling:** Services don't know about each other — they only emit and subscribe to events on the bus.
- **Extensibility:** New services or analyzers can be added by subscribing to existing events without modifying existing code.
- **Reliability:** If the dashboard goes down, honeypots continue operating and events queue in persistent storage.

3.2 Component Topology

The system comprises 10 interconnected modules organized in four layers:

MODULE	FILE	LINE S	FUNCTION
Orchestrator	server.js	150	Service lifecycle, logging, webhooks
SSH Honeypot	honeypots/ssh-honeypot.js	260	Real SSH protocol (ssh2), credential capture, interactive shell
HTTP Honeypot	honeypots/http-honeypot.js	318	Admin panels, sensitive file traps, attack detection
FTP Honeypot	honeypots/ftp-honeypot.js	165	ProFTPD emulation, directory traversal tracking
Telnet Honeypot	honeypots/telnet-honeypot.js	200	Interactive shell, Firecracker MicroVM isolation, fake responses
Attacker Profiler	intel/profiler.js	452	OS detection, MITRE mapping, behavioral scoring
IDS Engine	intel/ids-engine.js	78	Suricata EVE log integration
OSINT Engine	intel/osint-engine.js	93	VirusTotal, AbuseIPDB lookups
MicroVM Engine	honeypots/microvm-engine.js	150	Firecracker VM lifecycle, per-session isolation
Monitor Server	monitoring/backend/	480	REST API, WebSocket, auth, LevelDB

MODULE	FILE	LINE S	FUNCTION
	server.js		
PCAP Engine	monitoring/backend/pcap-engine.js	211	TCP session reconstruction, hex dump
Dashboard	monitoring/frontend/index.html	1505	10-page SPA, glassmorphism UI, reports

Total production code: ~4,100 lines (JS) + 1,505 lines (HTML/CSS/JS) = ~5,600 lines

3.3 Data Flow Pipeline

Attack events flow through the system in 6 stages:

- **Stage 1 — Capture:** Honeypot service receives attacker connection, parses protocol data, extracts credentials/commands/paths.
- **Stage 2 — Emit:** Service emits structured attack event on the EventEmitter bus with type, service, srcIP, details, analysis.
- **Stage 3 — Record:** Orchestrator logs to attacks.log, sends desktop notification, fires webhook. PCAP engine records packet data.
- **Stage 4 — Persist:** MonitorServer stores in LevelDB with UUID key. Optionally inserts into ClickHouse for analytics.
- **Stage 5 — Analyze:** AttackerProfiler updates the IP profile: OS guess, tools detected, TTPs mapped, behavior score recalculated.
- **Stage 6 — Push:** WebSocket broadcasts event to all connected dashboard clients for real-time display.

3.4 Technology Stack

COMPONENT	TECHNOLOGY
Runtime	Node.js 18+ (single-threaded event loop)
Framework	Express 4.21 (REST API)
Real-Time	ws 8.18 (WebSocket server)
Persistent Storage	LevelDB via level 8.0 (embedded key-value store)
Analytics DB	ClickHouse (columnar OLAP, optional Docker container)
GeoIP	geoip-lite 1.4 (MaxMind GeoLite2 database)
IDS Integration	Suricata EVE JSON log parsing

COMPONENT	TECHNOLOGY
Threat Intel APIs	VirusTotal API v3, AbuseIPDB API v2
WHOIS	ARIN RDAP REST API
Frontend	Vanilla JS SPA, CSS3 glassmorphism, particles
Deployment	Bash 4+, systemd, UFW/firewalld/iptables/nftables
TLS	Auto-generated self-signed certificate, custom cert support
Sandbox Isolation	AWS Firecracker MicroVM (per-session disposable VMs)
Containerization	Docker + Docker Compose (ClickHouse, Suricata)

CHAPTER 4: IMPLEMENTATION DETAILS

4.1 SSH Honeypot Service

Module: honeypots/ssh-honeypot.js (260 lines)

- **Real SSH Protocol (ssh2):** Uses the ssh2 library for genuine SSH key exchange, authentication negotiation, and channel management — attackers interact with a real SSH server, not a TCP socket.
- **Password Authentication Capture:** Intercepts SSH password authentication via the ssh2 'authentication' event, extracting exact username/password pairs with zero false positives.
- **Interactive Shell Session:** After 3 failed login attempts, grants shell access with PTY support, backspace/Ctrl-C handling, and realistic command responses — luring attackers deeper for intelligence.
- **Exec Channel Support:** Handles SSH exec requests (e.g., 'ssh user@host whoami') separately from interactive sessions, capturing one-shot commands.
- **TCP Fallback Mode:** Gracefully falls back to banner-only TCP mode if ssh2 library is unavailable, ensuring the service always starts.
- **PCAP Recording:** All packets recorded via PCAPEngine for post-session analysis with TCP handshake/teardown synthesis.

4.2 HTTP Honeypot Service

Module: honeypots/http-honeypot.js (318 lines)

- **Fake Login Panels:** Three convincing admin interfaces — generic Admin Panel, WordPress wp-login.php, and phpMyAdmin — with credential capture on form submission.
- **Sensitive File Traps:** Serves fake .env (with realistic DB credentials), wp-config.php.bak, backup.sql, .git/config, robots.txt with honeypot paths, and REST API endpoints.
- **Attack Detection (5 engines):** SQL injection (UNION, OR 1=1, DROP patterns), path traversal (../ sequences), XSS (<script>, onerror, javascript: patterns), command

injection (;, |, && with system commands), scanner detection (Nmap, Nikto, SQLMap, Masscan signatures).

- **Response Fidelity:** Sets Apache/2.4.56 Server header, PHP/8.1.2 X-Powered-By, and serves HTML with realistic PHP error patterns to maintain the deception.

4.3 FTP Honeypot Service

Module: honeypots/ftp-honeypot.js (165 lines)

- **ProFTPD Emulation:** Full FTP protocol implementation supporting USER, PASS, LIST, CWD, RETR, STOR, PWD, SYST, FEAT, PASV, PORT, TYPE, QUIT commands.
- **Deliberate Weakness:** Allows login after 3 failed attempts (configurable) — a deliberate design decision to lure attackers deeper into the honeypot for more intelligence gathering.
- **Directory Traversal Tracking:** Records all CWD attempts, detecting path traversal attacks (../../etc/passwd) and logging accessed paths.
- **File Operation Logging:** RETR and STOR commands logged as file_download and file_upload attack types for attacker behavior analysis.

4.4 Telnet Honeypot Service

Module: honeypots/telnet-honeypot.js (200 lines)

- **Interactive Shell:** State machine (login → password → shell) with realistic Telnet negotiation and BusyBox-style prompt.
- **Firecracker MicroVM Isolation:** Integrates with the MicroVM Engine (microvm-engine.js) to execute attacker commands inside disposable Firecracker microVMs — each session gets an isolated 128MB VM with its own rootfs copy, eliminating host compromise risk.
- **Graceful Fallback:** When Firecracker is unavailable, falls back to comprehensive fake responses for 30+ Linux commands (whoami, id, ls, cat, ifconfig, ps, netstat, env, etc.) with realistic output.
- **Command Capture:** Records all executed commands with timestamps for behavioral analysis and MITRE ATT&CK mapping.

- **Sensitive Data Traps:** Fake SSH private key, /etc/passwd with realistic user entries, env with fake AWS keys and database credentials to detect data exfiltration attempts.
- **Session Cleanup:** On disconnect, the MicroVM is killed and its rootfs copy wiped — zero forensic trace of attacker activity remains on the host.

4.5 Attacker Profiler Engine

Module: intel/profiler.js (452 lines)

The AttackerProfiler is the intelligence core of the system, building comprehensive profiles for each unique attacker IP:

OS Detection

- **User-Agent Analysis:** 24 regex patterns detecting Windows 7-11, macOS, Ubuntu, Debian, Fedora, CentOS, Red Hat, Arch, Kali, Parrot, Android, iOS, FreeBSD, OpenBSD.
- **SSH Client Fingerprinting:** 12 patterns identifying OpenSSH (Ubuntu/Debian variants), PuTTY, WinSCP, Paramiko, Bitvise, JSch, Dropbear, Windows OpenSSH.

Tool Detection (24+ tools)

Identifies attacker tooling from User-Agent strings and behavioral signatures:

CATEGORY	TOOLS DETECTED
Scanners	Nmap, Masscan, Nuclei, ZGrab, Censys, Shodan
Vulnerability	Nikto, SQLMap, WPScan, OWASP ZAP, Burp Suite
Brute Force	Hydra, Medusa
Frameworks	Metasploit Framework
Fuzzers	FFUF, DirBuster, Gobuster
Scripted	Python Requests/urllib, cURL, Wget, Go HTTP, Axios, Perl LWP, Scrapy

MITRE ATT&CK TTP Mapping

Maps attack events and commands to 18+ MITRE ATT&CK technique IDs:

TECHNIQUE ID	NAME	TRIGGER
T1110	Brute Force	credential_capture, login_attempt events

TECHNIQUE ID	NAME	TRIGGER
T1078	Valid Accounts	Successful credential use
T1059	Command and Scripting Interpreter	Shell command execution
T1083	File and Directory Discovery	Directory traversal, path traversal
T1005	Data from Local System	Sensitive file access
T1190	Exploit Public-Facing Application	SQL injection, command injection
T1033	System Owner/User Discovery	whoami, id commands
T1082	System Information Discovery	uname, env commands
T1003	OS Credential Dumping	cat /etc/passwd, /etc/shadow
T1016	System Network Configuration	ifconfig, ip addr commands
T1049	System Network Connections	netstat, ss commands
T1057	Process Discovery	ps aux command
T1105	Ingress Tool Transfer	wget, curl download commands
T1053	Scheduled Task/Job	crontab commands
T1222	File Permissions Modification	chmod +x commands
T1485	Data Destruction	rm -rf commands
T1562	Impair Defenses	iptables modification
T1552	Unsecured Credentials	history, .ssh/id_rsa access

Behavioral Scoring (0-100)

Multi-factor scoring algorithm weighing:

- Attack count (2 pts each, max 30)
- Credentials attempted (5 pts each)
- Commands executed (3 pts each)
- Services targeted (8 pts per service)
- Tools detected (10 pts per tool)
- TTPs mapped (4 pts per TTP)
- Sensitive paths accessed (7 pts each)
- Dangerous commands like wget, rm -rf, chmod (15-20 pts)
- Session duration bonuses (>5min: +10, >1hr: +15)

Threat classification: **LOW** (0-24) → **MEDIUM** (25-49) → **HIGH** (50-74) → **CRITICAL** (75-100)

Credential Pattern Analysis

Classifies attack strategies:

- **Password Spray:** 1 username with many passwords — testing common passwords against a known account
- **Username Enumeration:** Many usernames with 1 password — searching for valid accounts
- **Dictionary Attack:** Many usernames with many passwords — brute-force credential stuffing
- **Targeted:** Few usernames and passwords — specific account targeting

4.6 PCAP Synthesis Engine

Module: monitoring/backend/pcap-engine.js (211 lines)

- **TCP Reconstruction:** Synthesizes SYN → SYN-ACK → ACK handshake and FIN → ACK teardown for each session.
- **Hex Dump Visualization:** Wireshark-style hex dump with ASCII sidebar for packet inspection.
- **JSON Streaming:** Streams saved as JSON files in logs/pcap/ for dashboard consumption.
- **Connection Statistics:** Tracks total, successful, failed, and active connections across all services.

4.7 Monitoring Dashboard & API

Module: monitoring/backend/server.js (480 lines) + monitoring/frontend/index.html (1,505 lines)

REST API (20+ endpoints)

ENDPOINT	METHOD	FUNCTION
/api/status	GET	System status, uptime, service info
/api/attacks	GET	All attack events with pagination
/api/attack/:id	GET	Single attack detail
/api/credentials	GET	Captured credential pairs

ENDPOINT	METHOD	FUNCTION
/api/attackers	GET	Per-IP attack summaries
/api/services	GET	Per-service statistics
/api/timeline	GET	Hourly attack distribution
/api/intel/profiles	GET	All attacker intelligence profiles
/api/intel/profile/:ip	GET	Single IP deep profile
/api/report	GET	Full threat intelligence report
/api/pcap/streams	GET	All PCAP stream list
/api/pcap/stream/:id	GET	Single stream with packets
/api/export	GET	JSON/CSV data export
/api/reset	POST	Clear all data (danger zone)

Dashboard Pages (10 views)

- **Overview:** Threat level indicator, live stats, attack feed, service activity charts, top attackers
- **Attacks:** Full attack log with severity badges, filterable, clickable for detail modal
- **Services:** Per-service status cards with attack counts and port information
- **Credentials:** Captured username/password table with service and timestamp
- **Attackers:** Per-IP profile cards with geo, attack count, last seen
- **Intel:** Deep attacker profiles — OS, tools, TTPs, credential analysis, commands, paths, HTTP fingerprint, timeline
- **PCAP Streams:** Wireshark-style packet list with transcript and hex dump views
- **Analytics:** Severity breakdown bar, 24h attack timeline, top passwords/usernames/attack types charts
- **Reports:** Professional 9-section threat report with print-ready CSS formatting
- **Settings:** System configuration display, danger zone with data reset

4.8 Automated Deployment System

Module: scripts/deploy.sh (1,340 lines)

Comprehensive production deployment script supporting:

- **8+ Linux Distributions:** Ubuntu, Debian, Kali, CentOS, RHEL, Fedora, Arch, Alpine, openSUSE, Manjaro
- **Automatic Dependency Resolution:** Node.js installation via NodeSource → package manager → nvm → direct binary (4-level fallback)
- **npm Install Error Recovery:** Detects and handles permission errors, disk full, network failures, native build failures (auto-installs build-essential)
- **Port Conflict Detection:** Identifies blocking processes, distinguishes own previous instances
- **Firewall Auto-Configuration:** UFW, firewalld, iptables, nftables — detects and configures whichever is active
- **Systemd Service:** Auto-creates unit file with security hardening (NoNewPrivileges, ProtectSystem, MemoryMax, CPUQuota)
- **Backup and Rollback:** Auto-backup before every deployment, verified backup integrity, one-command rollback
- **Health Checks:** HTTP, API, WebSocket, and per-port verification with retry logic
- **Uninstall:** Complete removal with final backup, firewall rule cleanup, user removal

CHAPTER 5: TESTING AND RESULTS

5.1 Per-Service Test Results

SSH Honeypot

TEST CASE	INPUT	EXPECTED RESULT	STATUS
SSH Key Exchange	SSH client connects to port 2222	Real SSH handshake via ssh2 library	PASS
Password Auth Capture	SSH login with admin:password123	Exact credentials captured via auth event	PASS
Interactive Shell	3 failed logins, then commands	Shell granted, commands captured with responses	PASS
Exec Channel	ssh user@host 'whoami'	Command captured, response sent, channel closed	PASS
PCAP Recording	Multiple packets exchanged	JSON stream file created in logs/pcap/	PASS

HTTP Honeypot

TEST CASE	INPUT	EXPECTED RESULT	STATUS
Admin Panel Served	GET /admin	Login form with credential capture	PASS
Sensitive File Trap	GET /.env	Fake .env with DB credentials	PASS
SQLi Detection	GET /?id=1 OR 1=1	sql_injection event emitted	PASS
Path Traversal	GET ../../etc/passwd	path_traversal event emitted	PASS
Scanner Detection	Nikto User-Agent request	scanner_detected event, tool identified	PASS
Credential POST	POST /login with admin:test	credential_capture event emitted	PASS

FTP Honeypot

TEST CASE	INPUT	EXPECTED RESULT	STATUS
Banner Response	TCP connection to port 2121	220 ProFTPD 1.3.8 Server ready	PASS
Login After Failures	3 failed USER/PASS attempts	Login granted on 4th attempt	PASS
Directory Traversal	CWD ../../etc	directory_traversal event emitted	PASS
File Download Trap	RETR /etc/passwd	file_download event emitted	PASS

Telnet Honeypot

TEST CASE	INPUT	EXPECTED RESULT	STATUS
Shell Access	Login with admin:admin	Shell prompt displayed	PASS
Command Capture	whoami, ls, cat /etc/passwd	Commands logged, events emitted	PASS
Fake Responses	Various Linux commands	Realistic fake output returned	PASS
Session Timeout	2 minutes idle	Connection closed cleanly	PASS

5.2 Integrated System Test

Full Pipeline Result

A comprehensive test attack sequence targeting all 4 honeypot services was executed using generate_attacks.js. 68 attack events were captured across SSH, HTTP, FTP, and Telnet services. The AttackerProfiler assigned a behavior score of 100/100 (CRITICAL), identified 3 tools (cURL, Nikto, SQLMap), mapped 11 MITRE ATT&CK TTPs, and classified the credential pattern as dictionary_attack with 22 attempts across 3 usernames and 9 passwords. All events persisted in LevelDB, pushed via WebSocket to the dashboard in real-time, and the professional report generated successfully with all 9 sections populated.

5.3 Dashboard Verification

All 10 dashboard pages verified for correct rendering and data display:

PAGE	VERIFICATION	STATUS
Overview	Threat level, live stats, attack feed, charts render	PASS
Attacks	Full event log with severity badges, detail modal	PASS
Services	4 service cards with correct port/status	PASS
Credentials	Username/password pairs with service tags	PASS
Intel	Full profile: OS, tools, TTPs, commands, paths, timeline	PASS
Analytics	Severity bar, timeline chart, top lists	PASS
Reports	9-section report with print formatting	PASS
Settings	Config display, reset button functional	PASS

CHAPTER 6: SECURITY ASSESSMENT

A thorough security audit was conducted on the codebase. The following vulnerabilities were identified, assessed, and remediated:

6.1 Identified Vulnerabilities and Remediation

#	SEVERITY	VULNERABILITY	REMEDIATION	STATUS
1	CRITICAL	Auth bypass via X-Forwarded-For spoofing	_getIP() rewritten to use req.socket.remoteAddress only; X-Forwarded-For no longer trusted	FIXED
2	CRITICAL	Command injection in Telnet docker exec	Docker exec removed entirely; replaced with Firecracker MicroVM isolation + safe fake responses	FIXED
3	HIGH	Hardcoded credentials in source code	Credentials moved to environment variables; .env.example provided; docker-compose reads from .env file	FIXED
4	HIGH	No TLS on dashboard, API, or WebSocket	Auto-generated self-signed TLS certificate; HTTPS/WSS enabled by default; custom cert support via config/tls/	FIXED
5	HIGH	Unbounded memory growth	LRU eviction caps added: attacks (10K), credentials (5K), sessions (2K), connections (2K); 20% eviction on overflow	FIXED
6	HIGH	SSH honeypot using raw TCP instead of ssh2	Complete rewrite using ssh2 library; real SSH key exchange, password auth, interactive shell, exec channels	FIXED
7	MEDIUM	CSV injection in data export	Formula-character escaping added (=, +, -, @, tab, CR prefixed with single quote)	FIXED
8	MEDIUM	.env files not fully gitignored	.gitignore updated to cover *.env, monitoring/backend/.env, and config/tls/ directory	FIXED

Security Posture Summary

All 6 critical and high-severity vulnerabilities have been remediated. The auth bypass was the most dangerous — a single spoofed X-Forwarded-For header could grant full unauthenticated API access. The command injection in the Telnet honeypot could have led to host compromise. Both are now fully patched. The system now runs HTTPS by default, credentials are externalized, memory is bounded, and the SSH honeypot uses the real ssh2

protocol library.

6.2 Security Hardening Applied

- **Network Layer:** TLS encryption for all dashboard traffic. Self-signed certificate auto-generated on first run; production deployments can supply CA-signed certificates in `config/tls/`.
- **Authentication:** IP detection uses socket-level address only (`req.socket.remoteAddress`). X-Forwarded-For is no longer parsed, eliminating header spoofing attacks.
- **Credential Management:** All sensitive credentials (ClickHouse password, API tokens) read from environment variables. No secrets in source code. `.env.example` documents required variables.
- **Command Isolation:** Telnet shell commands execute inside disposable Firecracker microVMs (128MB RAM, 1 vCPU). Each session gets an isolated rootfs copy that is wiped on disconnect. Falls back to safe fake responses when Firecracker is unavailable.
- **Memory Protection:** In-memory arrays capped with automatic eviction (oldest 20% removed when limit reached). Prevents memory exhaustion under sustained attack.
- **Data Export:** CSV export sanitizes formula-injection characters (`=`, `+`, `-`, `@`) with single-quote prefix to prevent Excel formula execution.

CHAPTER 7: COMPARATIVE ANALYSIS

7.1 Feature Comparison Matrix

Y = Full support, P = Partial support, N = Not supported

FEATURE	OURS	COWRIE	T-POT	DIONAEA	OPENCANARY
SSH Honeypot	Y	Y	Y	N	P
HTTP Honeypot	Y	N	Y	Y	P
FTP Honeypot	Y	N	Y	Y	N
Telnet Honeypot	Y	Y	Y	N	N
Real-Time Dashboard	Y	N	Y	N	N
WebSocket Push	Y	N	P	N	N
MITRE ATT&CK Mapping	Y	N	P	N	N
Attacker OS Detection	Y	N	N	N	N
Tool Detection (24+)	Y	N	P	N	N
Behavioral Scoring	Y	N	N	N	N
Credential Analysis	Y	P	P	N	N
PCAP Reconstruction	Y	P	Y	P	N
Professional Reports	Y	N	P	N	N
One-Command Deploy	Y	P	Y	N	P
Rollback Support	Y	N	N	N	N
Lightweight (<5K LOC)	Y	P	N	P	Y

CHAPTER 8: CONCLUSION AND FUTURE WORK

8.1 Key Achievements

- **1. Multi-Protocol Deception:** Four interactive honeypot services covering SSH, HTTP, FTP, and Telnet — the most common attack vectors — in a single unified platform.
- **2. Deep Attacker Intelligence:** Automated OS detection (36 patterns), tool identification (24+ tools), MITRE ATT&CK mapping (18+ TTPs), and behavioral scoring (0-100) per attacker.
- **3. Real-Time Monitoring:** WebSocket-powered dashboard with 10 specialized views, glassmorphism UI, and professional report generation with print support.
- **4. Production Deployment:** 1,340-line deployment script supporting 8+ Linux distributions with 4-level Node.js fallback, firewall auto-config, systemd hardening, backup/rollback.
- **5. Compact Codebase:** ~5,600 lines of production code delivering feature parity with platforms that require 30+ Docker containers and 100K+ lines.
- **6. Security Hardened:** All critical vulnerabilities remediated — TLS encryption, auth bypass fixed, command injection eliminated via Firecracker isolation, credentials externalized, memory bounded.

8.2 Limitations

The following limitations are acknowledged:

- **Self-Signed TLS:** The auto-generated TLS certificate is self-signed. Production deployments should supply CA-signed certificates or use a reverse proxy with Let's Encrypt.
- **Medium Interaction:** Honeypot services emulate protocols at medium fidelity. Sophisticated attackers may detect the deception through timing analysis or missing protocol features.
- **Single Process:** Node.js single-threaded model limits throughput under extreme attack volumes. PM2 clustering would resolve this.

- **Firecracker Dependency:** Full command isolation requires Firecracker binary, kernel image, and rootfs. Falls back to fake responses when unavailable.

8.3 Future Directions

- **1. Let's Encrypt Integration:** Add ACME protocol support for automatic CA-signed certificate provisioning, replacing the self-signed default.
- **2. STIX/TAXII Integration:** Export threat intelligence in STIX 2.1 format for sharing with threat intelligence platforms.
- **3. Machine Learning:** Train anomaly detection models on attacker behavioral patterns for predictive threat scoring.
- **4. ForensiX Integration:** Feed captured malware samples from the honeypot directly into the ForensiX malware analysis pipeline.
- **5. SIEM Integration:** Export events in CEF/LEEF format for Splunk, QRadar, and ELK Stack ingestion.
- **6. DNS Honeypot:** Add DNS protocol emulation to detect DNS tunneling and domain enumeration.
- **7. Email (SMTP) Honeypot:** Capture phishing attempts and spam relay abuse.
- **8. High-Interaction Expansion:** Extend Firecracker microVM support to SSH and FTP honeypots for full high-interaction mode across all protocols.

8.4 Project Statistics

METRIC	VALUE
Total Lines of Code	~5,600 (Node.js + HTML/CSS/JS)
Backend Modules	11 (4 honeypots + MicroVM engine + 3 intel + 3 monitoring)
Frontend	1,505 lines (single-page application)
Dashboard Pages	10 specialized views
Protocols Emulated	4 (SSH, HTTP, FTP, Telnet)
MITRE ATT&CK Techniques	18+ technique IDs mapped
Tool Detection Signatures	24+ attacker tools
OS Detection Patterns	36 (24 UA + 12 SSH client)
REST API Endpoints	20+
Deployment Script	1,340 lines (8+ distros, 4-level fallback)
Dependencies	7 (express, ws, level, ssh2, geoip-lite, chalk, dotenv)

METRIC	VALUE
Language	100% JavaScript (Node.js)

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